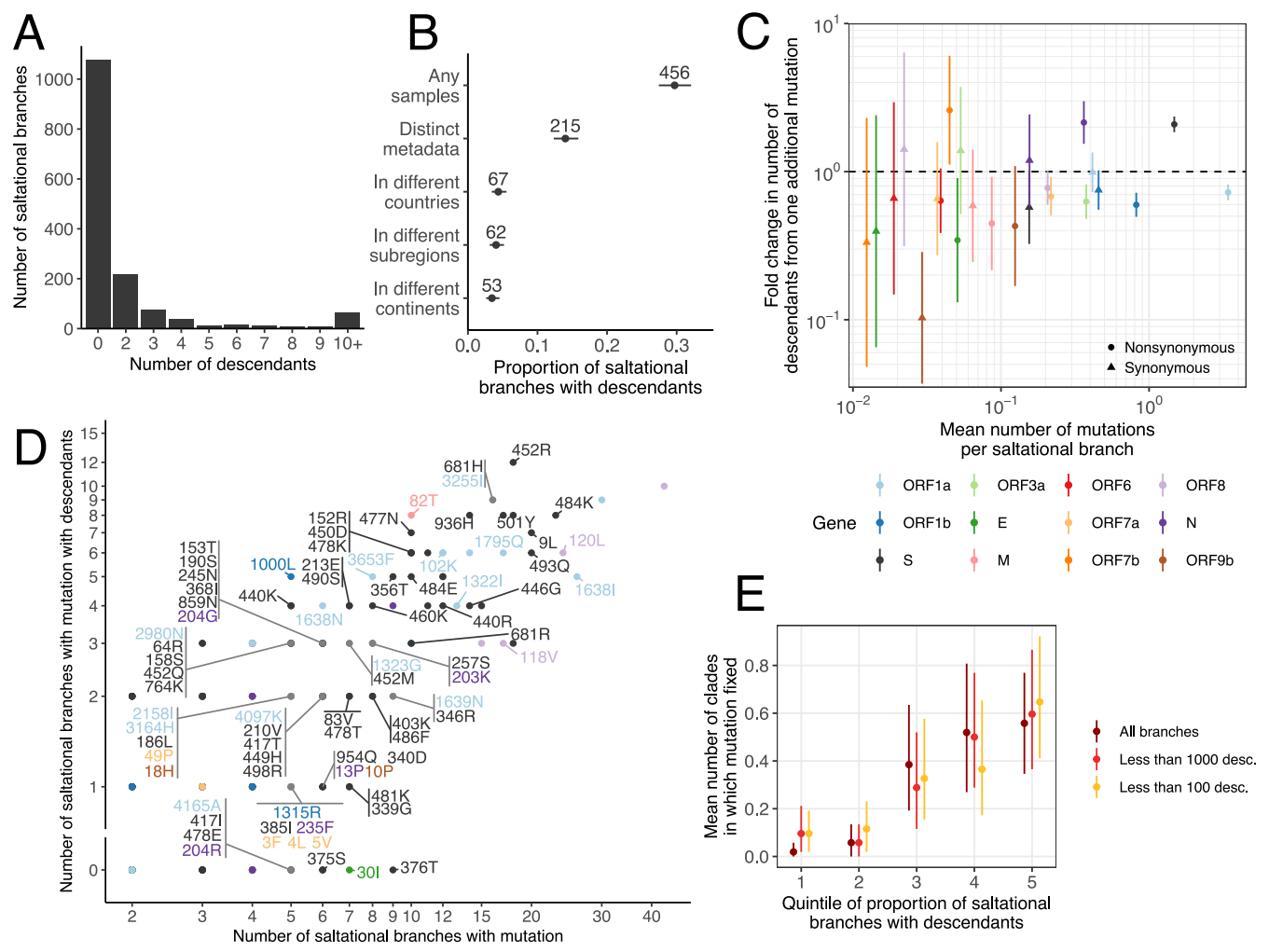

A
Number of saltational branches
1000
800
600
400
200
0
0 2 3 4 5 6 7 8 9 10+
Number of descendants
B
Any samples
Distinct metadata
In different countries
In different subregions
In different continents
456
215
67
62
53
0.0 0.1 0.2 0.3
Proportion of saltational branches with descendants
C
Fold change in number of descendants from one additional mutation
10^1
10^0
10^{-1}
Nonsynonymous
Synonymous
10^{-2}
10^{-1}
10^0
Mean number of mutations per saltational branch
Gene
ORF1a
ORF1b
S
ORF3a
E
M
ORF6
ORF7a
ORF7b
ORF8
N
ORF9b
D
Number of saltational branches with mutation with descendants
452R
681H
3255I
82T
484K
152R
450D
478K
477N
936H
501Y
1795Q
9L
120L
153T
190S
245N
368I
859N
204G
1000L
213E
490S
3653F
102K
493Q
356T
484E
1322I
446G
1638I
440K
1638N
460K
440R
681R
2980N
64R
158S
452Q
764K
118V
1323G
452M
257S
203K
1639N
346R
2158I
3164H
186L
49P
18H
4097K
210V
417T
449H
498R
83V
478T
403K
486F
954Q
13P
340D
10P
481K
339G
4165A
417I
478E
204R
1315R
385I
235F
3F
4L
5V
375S
30I
376T
2 3 4 5 6 7 8 9 10 12 15 20 30 40
Number of saltational branches with mutation
E
Mean number of clades in which mutation fixed
0.8
0.6
0.4
0.2
0.0
1 2 3 4 5
Quintile of proportion of saltational branches with descendants
All branches
Less than 1000 desc.
Less than 100 desc.